

NEUROSCIENCE OF EXECUTIVE FUNCTIONING

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EXECUTIVE FUNCTIONING FROM A LIFESPAN PERSPECTIVE

Although this chapter promotes a lifespan perspective on processes that are comprised in executive functioning (EF), the majority of research reviewed in the following sections relies on data from adults. This adult bias is inevitable, given that EF does not fully develop until adulthood and that, in children, components of EF are not sufficiently differentiated, either behaviorally or neuroanatomically (for a review, see Fiske & Holmboe, 2019). Research that does examine EF in children is primarily concerned with characterizing *developmental* trajectories of EF. This research has shown that rudimentary EF begins to emerge in infancy (approximately around 5–6 months of age) and experiences a fairly rapid development in early childhood (for reviews, see Best & Miller, 2010; Fiske & Holmboe, 2019). EF then continues to mature throughout adolescence and into the mid- to late 20s, as evidenced by improvements in both EF speed and EF accuracy during this period (Delis et al., 2001). Developmental improvements in performance parallel progressively greater processing efficiency (suggested by progressively more localized brain activation during EF task performance), as well as increases in synaptogenesis, neuronal proliferation and pruning, and myelination of relevant networks (for a review, see Best & Miller, 2010).

In adulthood, following 1 to 2 decades of stability, a gradual decline in the speed of EF performance begins to take place, starting in the mid-40s to early 50s (Delis et al., 2001). This initial decline may simply reflect general declines in speed of processing as it is not associated with increases in error rates. However, performance accuracy does begin to decline after age 60, with a continuing progressive decline thereafter (Delis et al., 2001). This decline is associated with more pronounced age-related cortical atrophy in prefrontal regions as compared with primary and secondary cortical areas, as well as declines in the efficiency of processing suggested by a more distributed (i.e., less localized) brain activation during performance of EF tasks (Greenwood, 2007). In summary, EF is the last neurocognitive domain to develop and among the first to begin to decline with age, peaking in young to middle adulthood.

EXECUTIVE FUNCTIONING FROM A HISTORICAL PERSPECTIVE

The notion that certain uniquely human qualities can be abolished with frontal lobe lesions began to be documented in the mid-19th century with the publication of several articles describing the famous case of Phineas Gage (Macmillan, 2000). Because no clear label existed at that time for the collection of abilities and processes that were impaired in patients

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with frontal lobe damage, by the mid-20th century these abilities came to be referred to as “frontal lobe functions” (Halstead, 1950). However, in subsequent decades, research increasingly demonstrated that not all frontal lesions lead to such deficits (Goodman & Kumar, 1992; Heck & Bryer, 1986; Hoffmann & Bar-On, 2012) and, conversely, that apparent “frontal lobe syndrome” was evident even among patients whose brain damage was restricted to nonfrontal structures (for a review, see Andrés, 2003). Thus, in the 1980s, the term “frontal lobe functions” began to be replaced with the term “executive functions”¹ (e.g., Goldberg et al., 1989; Lezak, 1982, 1983; Stuss & Benson, 1987), with the goal of arriving at a more functional, and less neuroanatomy-based, construct. Nevertheless, many researchers and clinicians have continued to use the two terms interchangeably to this day.

Aside from the difficulties in *naming* the construct, the field has also struggled with how exactly these functions should be organized or *conceptualized*. A variety of competing theoretical models of EF have emerged in the literature, some proposing a single underlying capacity and others proposing a multifaceted collection of neurocognitive processes (e.g., Baddeley & Della Sala, 1998; Lezak et al., 2004; Miyake et al., 2000; Norman & Shallice, 1986; Stuss, 2011; Suchy, 2015). These models were based on a range of methodologies, including functional fractionation of the presumed neuroanatomic substrates of EF, isolation of processes through experimental cognitive tasks, conceptual analyses of symptoms seen among patients with EF impairments, and factor analyses of available EF tests. These efforts culminated in the development of the so-called unity and diversity model of EF, which proposes that EF comprises three correlated but dissociable components: working memory, set shifting, and inhibition (Miyake et al., 2000).² Although this model was developed and validated factor analytically—an approach that has been criticized (Delis et al., 2003)—it has nevertheless come to dominate the field in recent years. Aside from the inherent tautology of the factor-analytic approach (i.e., “EF is

what EF tests measure”), the model is also necessarily constrained by the fact that it reflects only those processes that were entered into the model in the first place. Importantly, the processes that make up the model are based on conceptualizations that have been dominant among cognitive neuroscientists and fail to consider newly identified EF components that began to be recognized among clinical neuropsychologists in the 1990s.

In the 1990s, with the advent of advanced neuroimaging techniques, the focus of clinical practice and referral questions began to gradually shift from lesion localization to characterizing patients’ daily functioning (the presumed evolutionary purpose of EF). In the wake of this shift, it became increasingly clear that the available tests that were presumed to measure EF were failing to capture the full gamut of EF capacities needed for daily functioning. As a result, tests of EF began to be criticized as having poor ecological validity (Burgess et al., 1998). These criticisms instigated calls for the adoption of the so-called function-led approach to test development, that is, modeling new tests after EF applications in daily life (Burgess et al., 2006). Although this function-led approach to test development has had a strong following since its inception, it has not been matched by commensurate advances in EF conceptualization: In research and clinical practice alike, the EF construct is still all too often defined narrowly, either by neuroanatomy or by factor analyses of traditional EF tasks.

In this chapter, I advocate for a more comprehensive, function-led approach to defining the EF construct, one that is inspired by the function-led approach to test development (Burgess et al., 1998). This function-led EF construct deliberately encompasses capacities that are required for successful daily functioning, based on the assumption that the evolutionary purpose of EF is the ability to engage in adaptive, purposeful, and goal-directed behavior in daily life (Lezak et al., 2004; Stuss et al., 1997). Figure 8.1 provides an overview of the relevant historical developments and their correspondence to different EF conceptualizations.

¹Coined by Muriel Lezak (1982).

²This model has recently been shown to be unstable, with relatively little convergence among attempted replications (Karr et al., 2018).

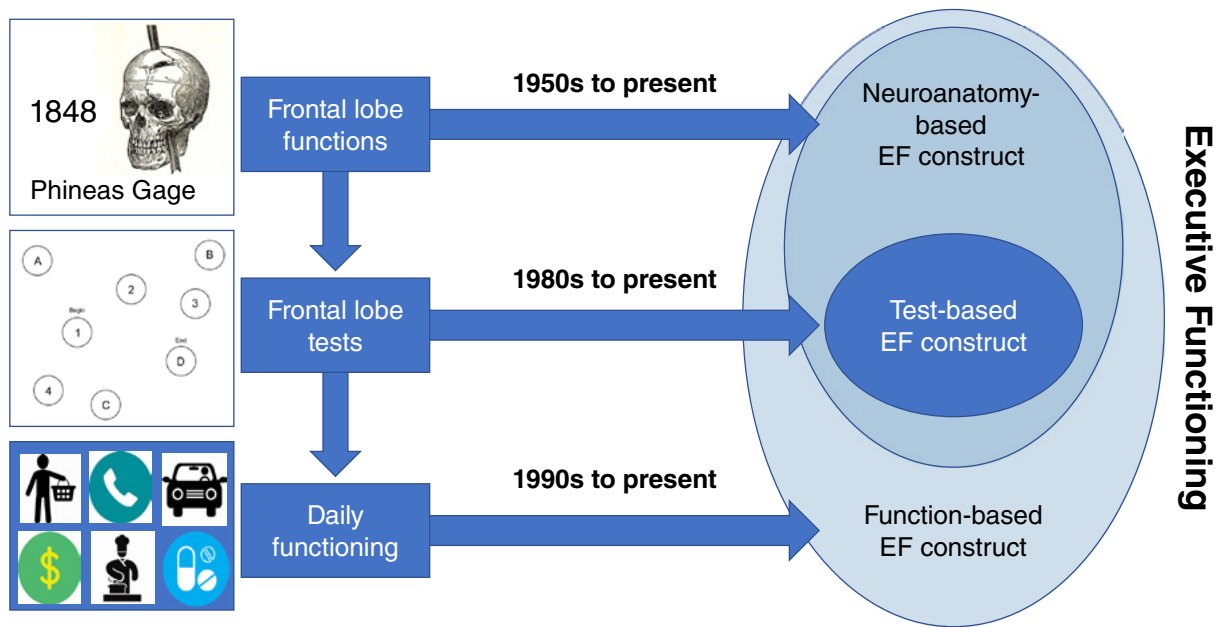


FIGURE 8.1. Evolution of the construct of executive functioning (EF) over the decades. Initially, the construct was defined exclusively by presentations of patients with frontal lobe lesions; over time, it shifted to being defined based on processes needed for performance of so-called frontal lobe tests; and finally, the construct has come to be viewed as a collection of processes that are needed for independent daily functioning. These shifts in the basis for defining EF have also led to changes in how broadly or narrowly the construct is defined. Importantly, all three bases for EF definition have persisted until the present, leading to considerable variability, both between and within disciplines, in how EF is conceptualized and measured.

WHAT THE FIELD KNOWS

This section reviews what the field knows about 12 elemental processes that have been extensively studied and that together make up the function-led EF construct. For convenience and ease of clinical translation, these processes are organized into four EF subdomains that correspond to four well-characterized clinical syndromes (see Volume 1, Chapter 9, this handbook). Additionally, each of the four subdomains is delineated by a cohesive collection of cognitive or behavioral outputs and by common neuroanatomic underpinnings. The four subdomains are referred to herein as the (a) executive cognitive functions, (b) initiation and maintenance of action, (c) response selection, and (d) metatasking. See Table 8.1 for an overview of these processes and associated neuroanatomic regions.

Executive Cognitive Functions

Executive cognitive functions (ECFs) refer to a set of higher order neurocognitive processes that allow

one to generate “cognitive products” that precede and are prerequisite to purposeful and goal-directed behavior. The cognitive products at hand are decisions about which goals to pursue (Del Missier et al., 2012), plans about how to effectively achieve the chosen goals, and solutions to problems when a given plan is thwarted. Because ECFs are responsible for adaptive goal selection and generation of plans *before* any action is taken, they do not, in and of themselves, guarantee that a plan will be executed or that a goal will be achieved (Shallice & Burgess, 1991).

Making a decision about which goal to pursue requires a process that is sometimes referred to as thoughtfully reflective decision making (TRDM), comprising (a) generation of several alternative choices or solutions; (b) collection of information about each choice; (c) systematic deliberation over alternatives, such as comparisons of pros and cons; and (d) retrieval of memories of prior decisions and comparisons of their outcomes with the current options (Paternoster & Pogarsky, 2009; Paternoster

TABLE 8.1

Overview of Elemental Processes and Neuroanatomic Regions Involved in Executive Functioning

Process/region	Executive cognitive functions	Initiation and maintenance	Response selection	Metatasking
Elemental processes	- Goal-directed search and retrieval - Working memory - Mental flexibility	- Effort mobilization - Initiation - Maintenance	- Discrepancy, error, or threat detection - Contingency updating - Behavioral inhibition	- Branching - Prospective memory - Metacognition
Lobes	- Frontal - Temporal - Parietal	- Frontal - Parietal	- Frontal - Temporal	- Frontal - Temporal - Parietal
Principal frontal convexity	- Dorsolateral^{a,b,c} - Ventrolateral (primarily left)^{a,c}	Medial^{d,e,f}	- Ventral/orbitofrontal^h - Right ventrolateralⁱ - Medial^g	Frontopolar^{j,k,l}
Discrete frontal cortical regions	- Dorsolateral PFC^{a,b} - Ventrolateral PFC^{a,c} - Cingulate gyrus (anterior and posterior)^b - Pre-SMA^c	- Superomedial PFC^{d,e,f} - Orbitofrontal/ventromedial^{d,f} - SMA^e - Pre-SMA^e - Anterior insula^f - Cingulate gyrus (anterior and middle)^{d,e}	- Orbitofrontal (ventromedial/ventrolateral)^h - Right inferior frontal gyrusⁱ - Anterior insula^g - SMA^g - Cingulate gyrus (anterior)^g	- Frontopolar^{j,k,l} - Dorsolateral PFC^l - Medial PFC^l
Discrete posterior cortical regions	- Posterior parietal cortex^a - Superior parietal lobule^{b,c} - Intraparietal sulcus^{b,c} - Medial temporal cortex	- Intraparietal sulcus^f - Temporoparietal junction^f	N/A	- Sensory areas^k - Cuneus/precuneus^k - Superior temporal gyrus^k
Subcortical and extracerebral regions	- Hippocampus^a - Cerebellum^b	- Ventral striatum^d - Cerebellum^f - Thalamus^f - Midbrain^f	Extended amygdala^g	Cerebellum^k
Relevant networks and circuits	- Frontoparietal (i.e., cognitive control or working memory)^{b,c} - Default mode^{a,c}	- Motivation^d - Reinforcement^d	- Salience^g - Error detection^g - Default mode^g	Default mode^l

Note. Involvement of frontal cortical regions assumes the involvement of the relevant frontal–basal ganglia–thalamic loops. The basal ganglia and the thalamus are only included in the table if they play a role beyond these loops. N/A = not applicable; PFC = prefrontal cortex; SMA = supplementary motor area.

^aGoal-directed search and retrieval. ^bWorking memory. ^cMental flexibility. ^dEffort mobilization. ^eInitiation. ^fMaintenance. ^gDiscrepancy, error, or threat detection. ^hContingency updating. ⁱBehavioral inhibition. ^jBranching. ^kProspective memory. ^lMetacognition.

et al., 2011). Although these four steps have been described as ingredients of decision making, they can be applied to planning (i.e., decisions about incremental subgoals) and to problem solving (i.e., decisions about alternative plans). Thus, though decision making, planning, and problem solving are sometimes described as distinct cognitive products,

they are, in reality, simply just different applications of the same set of underlying neurocognitive processes. These underlying processes include (a) goal-directed search and retrieval of relevant information, (b) manipulation of information in working memory, and (c) mental flexibility. These processes are described in turn next.

Goal-directed search and retrieval refers to a mental search for, and identification of, relevant semantic or episodic memories. Another well-recognized term that refers to a related construct is generative fluency. The role of this neurocognitive process in planning, reasoning, and problem solving has been well established (Brown et al., 2014; Schacter et al., 2015) and has to do with the requirement that relevant information be gathered before generation of plans or solutions. Because goal-directed search and retrieval relies in part on episodic memory, memory impairments necessarily preclude effective employment of ECFs (Addis et al., 2009; Volume 1, Chapter 10, this handbook). However, intact memory and learning capacity in and of itself is not sufficient for goal-directed search and retrieval because intact memory only requires that one be able to recall information on command. In contrast, goal-directed retrieval requires self-cuing and fluent generation of the targets of a given search.

Retrieval of information, whether semantic or episodic, relies in large part on the integrity of dorsolateral and ventrolateral convexities of the prefrontal cortex (PFC; Kostopoulos & Petrides, 2008; Suzuki et al., 2009), in conjunction with the posterior parietal cortex (Shimamura, 2011) and medial temporal structures (Gaesser et al., 2013). These neuroanatomic networks are known to play a role in performance on tests of verbal fluency (Costafreda et al., 2006; Verma & Howard, 2012), which are among the staples of neuropsychological assessment of EF (Rabin et al., 2005). Importantly, goal-directed retrieval differs neuroanatomically from simple retrieval on command, such that goal-directed retrieval relies on coactivation of memory retrieval networks with the default mode network (Gerlach et al., 2011), which itself has been implicated in the ability to recall and identify episodic memories that are relevant for imagining the future and, consequently, for prospective planning (K. C. R. Fox et al., 2015; Roberts et al., 2017).

The next elemental process of ECFs, *working memory* (WM), refers to the ability to hold and manipulate information in mind, and as such is key for the ability to mentally organize (e.g., sequence, compare) relevant pieces of information. The role of WM in planning, reasoning, and problem solving is

well established, so much so that some theoretical accounts consider WM to be a principal, or even the only, neurocognitive substrate of the EF construct (Baddeley & Della Sala, 1998; McCabe et al., 2010). Depending on the exact conceptualization, other terms that are synonymous (or nearly synonymous) with WM are cognitive or mental control. Consequently, the neuroanatomic network that subserves WM is sometimes referred to as the cognitive control network (K. C. R. Fox et al., 2015).

A large body of research, including single neuron recordings, functional imaging, and animal- and human-lesion studies, has shown that the dorsolateral PFC (DLPFC; particularly Brodmann's areas 46, 9, and 6) plays a key role in WM (Edin et al., 2009; Goldman-Rakic & Leung, 2002; Nácher et al., 2006). However, effective processing within these cortical areas intrinsically relies on intact cortical-subcortical loops that involve the dorsolateral head of the caudate nucleus, lateral dorsomedial globus pallidus internal, and ventral anterior and mediodorsal nuclei of the thalamus (i.e., the *direct* cortico-subcortical pathway), as well as dorsal globus pallidus external, lateral subthalamic nucleus, and globus pallidus/substantia nigra complex (the *indirect* pathways; Lichter & Cummings, 2001). Of note, the DLPFC appears to be specialized for content along a rostrocaudal gradient, such that the more abstract information is held in more rostral DLPFC regions, whereas the more concrete information is held in the more caudal DLPFC regions (Choi et al., 2018; Nee & Brown, 2012).

Effective WM additionally requires intact connectivity of the DLPFC with aspects of the parietal lobe, most notably, superior parietal lobule and intraparietal sulcus (Berryhill & Olson, 2008; Crone et al., 2006; Edin et al., 2009; Rottschy et al., 2012; Wendelken et al., 2008), the hippocampus (implicated in the retrieval of information from WM; Chein et al., 2011; Harms et al., 2013), and the anterior and posterior cingulate cortex (implicated in the maintenance of information in WM; Chein et al., 2011; Harms et al., 2013). WM also appears to rely on intact functioning of the cerebellum, especially the ventral regions of the dentate nucleus and associated networks, as evidenced by both functional imaging and lesion studies (Justus

& Ivry, 2001; Stoodley, 2012). Last, in light of the complexity of the WM networks, integrity of white matter tracts is also crucial for WM, in particular, the superior longitudinal fasciculus, the uncinate fasciculus, the cingulum, and the corpus callosum (Charlton et al., 2010; Østby et al., 2011; Treble et al., 2013).

Mental flexibility, the third aspect of ECFs, allows fluid mental travel between ideas, concepts, levels of analysis, perspectives, or perceptual sets. It is central to the ability to generalize from the past to the future or from the concrete to the abstract (Moscovitch & Winocur, 2002; Shimamura, 2002). An important aspect of mental flexibility is the notion of abandoning one mental set in favor of another. Thus, for example, when performing the Wisconsin Card Sorting Test (Heaton et al., 1993), an examinee must flexibly shift between *perceiving* color, form, or number of items presented on cards. Similarly, in many task-switching paradigms, fundamental mental and perceptual shifts are needed, such as switching between concrete versus abstract, color versus shape, or local versus global characteristics of a given stimulus (Cools et al., 2004; Huff et al., 2015; Stasenko et al., 2017). Of note, in cognitive neuroscience and clinical neuropsychology, the term “mental flexibility” (or “cognitive flexibility”) is often used somewhat loosely, referring to a broader set of processes, well beyond those described earlier. For example, “mental flexibility” is often used to describe the ability to adapt behaviorally or the ability to flexibly shift from one daily activity to another. However, these latter processes reflect behavioral, not cognitive, aspects of EF and rely on a separate set of neurocognitive processes and separate neuroanatomic networks that are discussed later in this chapter.

From a neurophysiologic standpoint, mental flexibility is thought to reflect the ability of discrete neuronal networks to become flexibly engaged or disengaged, a neuronal process known as oscillatory synchrony (Gregoriou et al., 2015). Of relevance here is the fact that imagining a stimulus and actually perceiving a stimulus activate the same neuronal network (Johnson & Johnson, 2014), and that imagery typically plays a role during planning and problem solving (Addis & Schacter, 2013; Madore

& Schacter, 2014). Neuroanatomically, the regions that subserve mental flexibility largely overlap with those that subserve WM, consistent with reported correlations between WM and switching abilities (Vandierendonck, 2012). Within the broader WM network, mental flexibility appears to be particularly related to the inferior frontal gyrus in the left hemisphere (Hirshorn & Thompson-Schill, 2006), pre-supplementary motor area (pre-SMA; Crone et al., 2006; Mansfield et al., 2011), and medial superior parietal lobule and intraparietal sulcus (Tamber-Rosenau et al., 2011), although the temporal lobes have also been implicated (Barrett et al., 2003; Smith et al., 2004). Lastly, mental flexibility, particularly as it relates to future-oriented thinking, also depends on coactivation between default mode network and WM networks (Roberts et al., 2017).

Initiation and Maintenance

Once goals are mentally selected and planned out, the actual attainment of goals depends on *initiation and maintenance* (I&M) of appropriate behavioral output. I&M is the first step in transitioning from the purely cognitive products of ECF to actual behavioral *execution*. I&M comprises (a) mobilization of effort, (b) purposeful and internally motivated initiation of action (which is in contrast to reflexive, automatic, externally motivated actions), and (c) continuous refreshing of one’s mental set so as to maintain relevant behavioral output. It is important to dissociate I&M from ECFs, such that intact ECFs do not guarantee that relevant actions will be initiated and followed through. Conversely, deficits in ECFs do not preclude normal employment of I&M—in other words, regardless of whether a plan is generated by one’s own ECFs or whether it is provided by another person, I&M can proceed.

The first elemental process of I&M, *effort mobilization*, has also been referred to as “energization” (Stuss, 2011), “psychomotor” or “incentive sensitization” (Robinson & Berridge, 2000), “behavioral activation” (Salamone et al., 2007), or “wanting” (Berridge & Kringelbach, 2008). It has been conceptualized by a number of authors as an aspect of EF (for a review, see Ducharme et al., 2018) and refers to an inherent tendency to physically engage with the environment for the purposes of either goal

attainment or, in the absence of an explicit goal, environment exploration. Although effort mobilization is a prerequisite to most any action (Desmurget, 2013), it relies in part on autonomic activation (Silvia et al., 2014) and as such must strike a balance between the availability of physiologic resources and the inherent tendency to conserve such resources (Silvestrini & Gendolla, 2013). The neuroanatomic networks that subserve effort mobilization include the anterior and middle cingulate cortex and medial superior frontal gyrus and the associated basal ganglia circuits (Levy & Dubois, 2006), along with motivational (i.e., reinforcing) contributions from the orbitofrontal and ventromedial cortices and ventral striatum (Massimo et al., 2009; Schlund et al., 2012; for a review, see Ducharme et al., 2018).

Initiation, the second elemental process of I&M, refers to processing that immediately precedes observable motor output but is also inextricably a part of that output. This process involves an assembly of discrete components of motor or verbal sequences. Neuronal activation that immediately precedes movement involves multiple motor circuits within the cortex, the basal ganglia, and the pyramidal tracts. For reviews, see Klaus et al. (2019) and Volume 1, Chapter 8, this handbook. Although initiation is sometimes conceptualized as motor or action planning, it is nevertheless a cognitive process that forms a bridge between processes that are purely motor and those that reflect higher order cognition. For example, action-planning latencies (i.e., a brief pause before an action is released) evidence strong associations (i.e., large effects sizes) with performance on measures of EF (Niermeyer et al., 2017; Suchy & Kraybill, 2007; Suchy et al., 2010), as well as with the magnitude of the so-called motor readiness potential, which itself is also correlated with EF (Euler et al., 2015). Impairments in initiation are characterized by longer latencies preceding motor or verbal output, as well as assembly of shorter sequences of action components, which ultimately result in slower, less smooth, and less efficient performance overall (Verwey et al., 2011). Initiation relies on superomedial aspects of the frontal lobes, including the anterior cingulate gyrus, rostral medial PFC, supplementary

motor area (SMA), and pre-SMA (Hoffstaedter et al., 2014; Massimo et al., 2015; Volle et al., 2012).

The last aspect of I&M is *maintenance*. It refers to an elemental process responsible for maintaining intended behavioral output over time. Maintenance can be conceptualized as the capacity for vigilant attention. There is considerable evidence that the ability to maintain attention relies on the ability to continually refresh a certain configuration of neuronal firing (or oscillatory synchrony), with each refresh cycle lasting approximately 10 seconds (for a review, see Langner & Eickhoff, 2013). The neural network that controls the refresh cycles is largely right-hemisphere dominant and includes the dorsomedial, medial, and ventromedial PFCs; anterior insula; intraparietal sulcus; temporoparietal junction; cerebellar vermis; thalamus; putamen; and midbrain (Langner & Eickhoff, 2013).

Response Selection

Response selection (RS), the third subdomain of EF, comes into play once plans are translated into actions. RS refers to rapid and immediate behavioral decisions that become necessary when contextual changes signal that ongoing behavior is no longer adaptive or that new options are available. In such situations, a rapid decision needs to be made about whether to exploit the ongoing behavior or explore new behavioral opportunities (Domenech & Koechlin, 2015; Donoso et al., 2014). Contextual changes and availability of options are ever present, so all behavior must be accompanied by sustained monitoring for potentially relevant cues (e.g., a change from green to red on a traffic light). As was the case with I&M, the products of RS are inherently “behavioral” in that any processes that precede a given action are inextricably linked to that action and, if separated from the behavioral outcomes, no longer hold any relevance. Therefore, the rapid behavioral decisions should not be confused with deliberative decision making. In fact, many daily behavioral choices do not afford adequate time for deliberation (Paternoster & Pogarsky, 2009), requiring instead that a decision be made on the fly. Ultimately, the goal of such rapid decision making is to maximize rewards and minimize punishments in the here and now.

Different theoretical models for such rapid decision making have been proposed. Among the most prominent is the somatic marker hypothesis, which proposes that rapid decisions are based on gut feelings that reflect physiological changes associated with certain situations and stimuli (Bechara et al., 2005). In brief, according to this model, the brain stores memories of autonomic experiences associated with prior outcomes of specific behavioral choices. Thus, whenever the same behavior is initiated, physiologic memories signal whether the expected outcome is likely to be positive or negative. This model incorporates cognitive, emotional, and physiologic processes and explains the rapid decision-making process in terms that are highly intuitive and match the subjective phenomenology of the human experience; it therefore has gained a considerable popularity and prominence in the literature. In sharp contrast to the somatic marker hypothesis is a computational model that is based on the Dirichlet process mixture models, a type of calculation used in Bayesian statistics (Donoso et al., 2014). This model, while conceptually and mathematically complex, allows for elegant comparison of actual behavioral responses to responses modeled mathematically and thus illustrates the feasibility of making somewhat complex adaptive choices in the absence of human consciousness. Briefly, this model proposes two inferential streams within the PFC: one that makes predictions about the certainty of an ongoing behavior, and one that makes predictions about the certainty of alternative behavioral options. Despite the apparent differences between the somatic marker hypothesis and the Dirichlet process model, they share the notion that neurocognitive processes that enable RS occur largely on a preconscious level, involving (a) discrepancy, error, or threat detection; (b) rapid and flexible updating of previously learned contingencies; and (c) inhibition of one response in favor of another response (for a review, see also Castellanos-Ryan & Séguin, 2016).

The first aspect of RS, *detection of discrepancies, errors, or threats*, refers to a continuous monitoring of one's actions and external context for

discrepancies with plans or expectations. This monitoring occurs automatically and on a preconscious level, as suggested by electroencephalography research showing that error-related negativity occurs before awareness (O'Connell et al., 2007; Orr & Hester, 2012). At least two separate brain systems contribute to the detection of discrepancies, residing in medial frontal and frontotemporal regions, largely within the right hemisphere (Rolls, 2004; Ullsperger et al., 2010). The first system, sometimes referred to as the *salience network* (Menon & Uddin, 2010), is primarily concerned with detection of relevant sensory information. It is composed of a population of von Economo and fork neurons, which are bipolar neurons that connect the right anterior insula, the anterior cingulate gyrus, the inferior frontal gyrus, and the amygdala (Allman et al., 2010; Menon & Uddin, 2010). According to some theoretical accounts, the anterior insula within this network serves as a hub that connects systems that attend to external stimuli (e.g., the extended amygdala, the WM network) with the internally oriented default mode network³ (Menon & Uddin, 2010). Additionally, the extended amygdala is well known for its sensitivity to threatening and socially relevant stimuli and its role in triggering autonomic cascades associated with fight-or-flight response (A. S. Fox et al., 2015). On the whole, the salience network plays a role in detecting stimuli that signal the need for alteration in ongoing behavior.

The second system, typically referred to as the *error-detection system*, involves the anterior cingulate gyrus and SMA (Ridderinkhof et al., 2004) and has not only the capacity to rapidly and preconsciously identify errors but also the capacity to rapidly code their significance based on context (Maier et al., 2012). Error monitoring relies on task rules that are held in the lateral PFC (Zarr & Brown, 2016). Both task rules within the lateral PFC and error monitoring/prediction within the medial PFC appear to be coded along a rostrocaudal gradient, such that more rostral regions hold more abstract task rules and monitor for more abstract errors, whereas more caudal regions hold more concrete task rules and monitor for more concrete errors (Zarr & Brown,

³The default mode network comprises the medial PFC, posterior cingulate cortex, precuneus, and angular gyrus. The default mode network is activated during mind wandering or daydreaming, such as during resting-state functional imaging paradigms.

2016). Importantly, error detection is associated with an autonomic cascade including an increase in cortisol, which also appears to contribute to behavioral slowing, thereby facilitating behavioral change (Tops & Boksem, 2011). Of note, consistent with the somatic marker hypothesis, a conscious awareness of errors or discrepancies is associated with awareness of autonomic responses, coded by the right anterior insula (Ullsperger et al., 2010).

The second aspect of RS, *contingency updating*, refers to a rapid, preconscious updating of previously learned associations between behaviors and outcomes, considering a broad array of cues that signal potential threats or rewards and thus require targeted allocation of attentional resources (Cox et al., 2005; Grossberg et al., 2008). Contingency updating allows one to change or reverse a response to a given stimulus or, conversely, to change which stimulus is used as a trigger for a particular response. A classic example from clinical neuropsychology is the switches that are built into the administration of the Wisconsin Card Sorting Test (Heaton et al., 1993). Similar to the capacity of the error-detection system to implicitly code the significance of errors (Maier et al., 2012), the contingency-updating system has the capacity to implicitly compute the probabilities of positive or negative outcomes and to conduct a rudimentary cost–benefit analysis (Rolls, 2004; Windmann et al., 2006).

The primary substrate of contingency updating is the orbitofrontal cortex, including ventrolateral and ventromedial regions (Rolls, 2004). Within these regions, the coding of more abstract (e.g., monetary) and more temporally distant rewards is thought to occur in the more rostral regions, whereas more concrete (e.g., food) and more immediate rewards are thought to be localized caudally (Kringelbach & Rolls, 2004). There is less agreement with respect to coding of valence, with some accounts suggesting that coding of rewards and punishments takes place in the medial and lateral orbitofrontal cortices, respectively (Grabenhorst et al., 2007; Ursu & Carter, 2005), and others

suggesting that punishments are processed in the right ventromedial orbitofrontal cortex (Clark et al., 2003; Tranel et al., 2002; for reviews, see Rolls, 2004, and Windmann et al., 2006).

The last elemental process of RS, *inhibition*, refers to the stopping and prevention of motor actions. Inhibition can occur in a top-down fashion in response to a deliberative decision making, or it can be triggered by bottom-up processes such as discrepancy detection and contingency updating and can take place without a conscious awareness (Hughes et al., 2009). Because the neural processing of inhibition of a response is faster than response initiation (i.e., 200 vs. 250 ms, respectively), it can be effected even *after* a given motor program has been released and abort an incipient action (Logan & Irwin, 2000). For a review, see Rossetti and Pisella (2003) and Volume 1, Chapter 8, this handbook. Response inhibition relies primarily on the right inferior frontal gyrus (Aron et al., 2014; Chamberlain & Sahakian, 2007; Chikazoe, 2010), which coordinates the inhibitory action with other brain regions involved in discrepancy detection (i.e., the right orbitofrontal gyrus, the anterior cingulate gyrus, pre-SMA) as well as with those involved in generation of plans for future actions (i.e., the WM networks; Cohen et al., 2013; Roger et al., 2014).

Metatasking

Metatasking (MT) is the final subdomain of EF⁴

It refers to one's ability to effectively interleave subgoals of one task with subgoals of other tasks, allowing for a completion of multiple, *multitask* tasks across somewhat extended periods of time. MT is not possible without intact ECFs, I&M, and RS; however, these subdomains, while necessary, are not sufficient for MT to take place. This section focuses on those elemental aspects of MT that are distinct from those already covered under the previous three EF subdomains. Specifically, MT comprises three elemental neurocognitive processes, including branching, prospective memory, and metacognition (Logie et al., 2010; McAlister & Schmitter-Edgecombe, 2013).

⁴Although the term “multitasking” is sometimes used to refer to the same construct, that term is also used to describe other activities, such as rapid switching between two simple tasks or completing two simple tasks at once (e.g., the proverbial walking and chewing gum). To avoid confusion, the term “multitasking” is not used in this text.

Branching refers to the ability to temporarily abort performance of one task for the purpose of completing another task and then return to the original task as appropriate. Thus, branching requires the ability to hold in mind multiple tasks and goals and flexibly switch among them, requiring initiation, stopping, and reinitiation of activities, often in the absence of any external cues. In clinical neuropsychology, the Condition 4 of the Color–Word Interference Test from the Delis–Kaplan Executive Function System battery (Delis et al., 2001) assesses branching, requiring that examinees name the color of ink (i.e., Stroop interference effect), except if the word is placed in a box, in which case they should simply read the word. Research examining the neuroanatomy of branching has consistently implicated the frontopolar cortex (Koechlin et al., 1999; Mansouri et al., 2017).

Prospective memory refers to the capacity to retrieve prior intentions in response to external cues (event-based prospective memory) or passage of a certain amount of time (time-based prospective memory). Prospective memory requires not only the ability to remember what one's intentions are (clearly requiring intact episodic memory) but also the ability to retrieve such intentions when appropriate, which typically requires strategic monitoring of the environment for relevant cues or strategic monitoring of the passage of time. There is some evidence that event-based and time-based prospective memories are dissociable, such that tasks of time-based prospective memory have consistently been found to be more difficult than tasks of event-based prospective memory (Gonneaud et al., 2014). Additionally, although functional imaging and lesions studies show that both time-based and event-based prospective memories rely on the integrity of the frontopolar cortex (Burgess et al., 2011; Volle et al., 2011), the two types of prospective memory differ with respect to the involvement of additional brain regions. Specifically, event-based prospective memory is associated with additional activation in relevant sensory areas, likely due to monitoring the environment for cues, whereas time-based prospective memory involves additional extensive activation of the WM networks, the cuneus and precuneus, superior temporal gyrus, and cerebellum (Gonneaud et al., 2014).

The third elemental process of MT is *metacognition*, sometimes also referred to as insight or self-knowledge. Metacognition has been shown to be related to the ability to metatask in daily life (Fazeli et al., 2017) and to performance on EF tests (Fazeli et al., 2017; Villalobos et al., 2021). It reflects one's capacity to retrospectively judge how well one accomplished a given task and to prospectively estimate how well one will be able to accomplish an ongoing or future task (Fleming & Dolan, 2014). With respect to achievement of MT goals, metacognition allows us to judge (a) which subgoals have been completed and which have yet to be done, (b) whether subgoal completion is accurate or matches a plan, and (c) whether our own cognitive resources (e.g., episodic memory, WM) are sufficient for successful execution of MT activities or whether compensatory strategies such as reminders, written checklists, and the like need to be employed. Various frontal lobe structures have been implicated in metacognition using functional (De Martino et al., 2013; Paul et al., 2015; Valk et al., 2016) and structural (Grossner et al., 2018) imaging research, with some research pointing specifically to the frontopolar cortex (Miele et al., 2011). Additionally, there appears to be some distinction between retrospective versus prospective metacognitive judgments, such that the lateral frontopolar and dorsolateral cortices play a role in retrospective judgments carried forward when planning an execution of a new task, whereas medial structures (including the default mode network; Grossner et al., 2019) play a role in the retrospective and immediate judgments (for a review, see Fleming & Dolan, 2014). However, findings regarding additional structures implicated by functional imaging research are quite varied, perhaps because of differences in the specific content of a given metacognitive task (Baird et al., 2015; Valk et al., 2016).

Summary and Conclusions

In this chapter, EF is defined as a comprehensive construct responsible for selection, planning, and execution of future-oriented behavior in daily life. The 12 neurocognitive processes that fall under this comprehensive umbrella can be organized into four subdomains: executive cognitive functions, initiation and maintenance, response selection, and

metatasking. Each subdomain comprises processes that share (a) overlapping neuroanatomic networks and (b) conceptually related behavioral outcomes. In addition, deficiencies in each of the four subdomains correspond to the previously described four neurobehavioral syndromes (see Volume 1, Chapter 9, this handbook). From a neuroanatomic standpoint, each of the four subdomains relies in part on frontal lobe structures, consistent with early observation of EF deficits following frontal lobe lesions. However, the regions and networks that are needed for intact EF extend well beyond the frontal lobes, also including the frontoparietal network (also known as the cognitive control network), default mode network, and salience network. Together, these networks involve both cerebral hemispheres and encompass frontal, parietal, and temporal structures, as well as the diencephalon and the cerebellum, making EF truly a whole-brain process. See Table 8.1 for an overview.

ACCEPTED SCIENCE

- Executive functioning (EF) is multifaceted, comprising multiple elemental processes.
- EF is subserved by the frontal lobes in conjunction with regions within the parietal and temporal lobes, cortico-subcortical loops (including the basal ganglia and the thalamus), default mode network, and cerebellum.
- Four of the 12 neurocognitive processes described in this chapter (working memory/cognitive control, generative fluency, mental flexibility, and inhibition) are virtually universally accepted as aspects of EF; the remaining eight are included in some, but not all, theoretical models of the construct.

QUESTIONS AND CONTROVERSIES

As already alluded to in the introduction to this chapter, arguably the most prominent model of EF is the one proposed by Miyake and colleagues (2000). The model proposes that EF comprises three related entities: working memory, set shifting, and inhibition. The model was developed factor

analytically and has been slightly revised over the years to accommodate the emergence of new factor-analytic findings (for a review, see Karr et al., 2018). Aside from the well-described limitations of factor analysis as a methodology for investigation of neurocognitive constructs (Delis et al., 2003), the Miyake model is limited in that it does not clearly differentiate conceptually (and sometimes methodologically) between cognitive and behavioral processes. For example, *cognitive inhibition* (i.e., keeping unwanted percepts or thoughts from entering working memory) is confounded with *behavioral inhibition* (i.e., stopping a behavioral response). Similarly, *mental flexibility* (i.e., the ability to change one's mental set) is confounded with *behavioral flexibility* (i.e., the ability to update reinforcing contingencies toward a behavioral change).

Such semantic overlap among functionally and neuroanatomically distinct constructs (as reviewed earlier in this chapter) has represented an insidious but persistent barrier to progress in investigations of EF, making it difficult for studies or models to build upon each other. This barrier is further augmented by the pervasive disconnect between conceptualizations of EF that are strictly cognitive in nature and those that take a broader behavioral perspective. The former approach is typically adopted in cognitive psychology and cognitive neuroscience and closely links conceptualizations of EF to available cognitive paradigms, typically conducted with healthy participants. The latter approach incorporates experience with patient populations and, in addition to test performance, considers functioning in daily life. Although there is some permeability between the two approaches, pervasive limitations in semantics and terminology preclude fruitful “cross-pollination.” Thus, it can be argued that the most pressing issue in the study of EF today is the need for an interdisciplinary consensus about definitions and terminology pertaining to discrete elemental processes. Additionally, consensus about the scope of the EF construct is needed, as it has been argued that a thorough understanding of EF requires that both cognitive and behavioral processes be included under the umbrella of this complex construct (Ardila, 2008).

QUESTIONS AND CONTROVERSIES SUMMARY

- Preconscious processes, such as sensitivity to discrepancies or threats, have not been traditionally considered to be aspects of executive functioning (EF). However, as described in this chapter, such processes appear necessary for proper response selection (RS). Would assessment of such processes contribute independently to assessment of EF, or are deficiencies in such processes captured “downstream” by assessment of other aspects of RS?
- Do discrete deficits in individual elemental processes affect the entire EF subdomain, or is there a dose effect, such that a greater number of impaired processes within a subdomain would be associated with a greater functional impairment?

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